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PROJECT REPORT

ON

“IMAGE CAPTIONING USING DEEP LEARNING”

Submitted In Partial Fulfillment of The Requirements

For

VI SEMESTER

BACHELOR OF COMPUTER APPLICATIONS

SUBMITTED BY

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2024 – 25

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CERTIFICATE

This is to certify that the project entitled “**Image Captioning Using Deep Learning**” has been successfully submitted by **SHAIKH SUFFI NAAZ** RegNo. **U23IA22S0069**.

A Bonafide student at Government Degree College, Yadgir in partial fulfillment of award of Bachelor of Computer Applications in the year 2024 – 2025 is record of student own work carried out under my/our guidance. It is certified that all corrections/suggestion indicated for internal assessment have been incorporated in the report & one copy of it being deposited in the Government Post-Graduation Library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed for the said bachelor’s degree. It is further understood that this certified the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion draw there in but approved the project only for the purpose for which it is submitted.

Examiners:-

1.

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Guide
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ACKNOWLEDGEMENT

I express my sincere gratitude to my guide, **Shri. Manjunath. V, Assistant Professor** of department of computer applications to guide for their guidance, proper advice & encouragement during the course or work.

I also feel very obliged to **Dr. Basavaprasad. B, Head of the Department** of computer applications for his encouragement & inspiration for execution of this thesis work.

I also feel thankful to **Dr. K Subhashchandra, Principal** of Government Degree College, Yadgir Support during work.

I am also thankful to the entire faculty & staff members and my friends from the computer applications department for their direct & indirect help & cooperation.

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CANDIDATE DECLARATION

SHAIKH SUFFI NAAZ a student of Bachelor's in Computer Applications department bearing Regno **U23IA22S0069** of **Government Degree College, Yadgir** hereby declare that my full own responsibility for the information, results and conclusions provided in this work titled **“Image Captioning Using Deep Learning”** submitted to **Adikavi Maharishi Valmiki University, Raichur** for the award of Bachelor's Degree In Computer Applications.

To the best of my knowledge, this project has not been submitted in part or full elsewhere in any other institution/organization for the award of any certificate/diploma/degree. I have completely taken care of acknowledging the contribution of others in academic work. I further declare that in any case the candidate will be solely responsible for the same.

Date:-

Place:-

Signature of the candidate

Name:- SHAIKH SUFFI NAAZ

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Abstract

Image captioning is a challenging and significant task in artificial intelligence that combines computer vision and natural language processing to generate meaningful textual descriptions for images. This project aims to develop an automated image captioning system using deep learning techniques, integrating a **Convolutional Neural Network (CNN)** for visual feature extraction and a **Recurrent Neural Network (RNN)**, specifically **Long Short-Term Memory (LSTM)**, for text generation.

The model follows an **encoder-decoder architecture**, where the encoder (a pre-trained CNN, such as **ResNet-50, InceptionV3, or VGG16**) extracts high-level image features, which are then passed to the decoder (LSTM) to generate coherent captions. Enhance accuracy, an **attention mechanism** is incorporated, allowing the model to focus on specific image regions while generating words sequentially. The training process utilizes a large dataset, such as the **MS COCO dataset**, which contains thousands of images paired with human-annotated captions. The dataset is pre-processed through image resizing, tokenization of captions, and word embedding techniques using **GloVe or Word2Vec**.

Improve the quality and relevance of generated captions, **transfer learning** is applied to the CNN, and **beam search decoding** is used instead of traditional greedy search to generate better sentence structures. The performance of the model is evaluated using standard **natural language generation metrics**, including **BLEU, METEOR, ROUGE, and CIDEr scores**, which measure the similarity between generated captions and ground-truth human descriptions.

The proposed system has a wide range of applications, including **image-based search engines, automatic content generation, assisting visually impaired users with real-time image descriptions, and digital media enhancement**. The results indicate that deep learning-based image captioning significantly improves the understanding and representation of visual content, bridging the gap between image recognition and language modelling. This project demonstrates the potential of deep learning in achieving human-like interpretation of visual data, contributing to advancements in AI-powered communication and accessibility technologies.

Keywords: -

CNN, RNN, LSTM, Encoder-Decoder, ResNet 50, Ms Coco Dataset, GloVe, BLEU, Meteor, Rouge, Cider.

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